

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

May 2014

EC 407 Radar Systems

Date: 24.05.2014, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What is the role of Loop Antenna in Target Direction finding in Radar System? [03]
Explain Briefly.

(b) 1) How to increase Sensitivity of Direction Finder in Radar System? [03]

2) List Out the Navigation Parameters. [01]

Q - 2 (a) What is the Role of Mixer in Radar Explain Briefly? [04]

(b) Draw The Block Diagram of Radar and Explain all the Blocks in details. [06]

(c) What is the application of Radar? Explain it briefly. [04]

OR

Q - 2 (a) Give Insight for Propagation Loss Calculation in Radar System. [03]

(b) Explain Following Terms briefly. [08]

(a) Pulse Repetition Frequency (b) Doppler Effect

(c) Noise Figure (d) Duplexers

(c) Explain Radar Equation and its Significance. [03]

Q - 3 (a) Explain Range gate Doppler Filters. [05]

(b) Explain Delay line Canceller in MTI Radar. [05]

(c) Explain How MTI can detect the target from moving platform. [04]

OR

Q - 3 (a) What are Limitation of MTI ? [04]

(b) Explain Autocorrelation Function and its Significance in radar receiving System? [05]

(c) Explain Multiple or Staggered radar system. [05]

SECTION – II

- Q - 4 (a) Explain LORAN-A system in Detail. [04]
(b) Explain Silent Features Of TACAN. [03]
- Q - 5 (a) Derive the Doppler Range equation for Doppler Navigation. [07]
(b) Draw the block diagram of FMCW altimeter and explain it. [07]

OR

- Q - 5 (a) Draw neat block diagram of FMCW radar and enlist its distinguished features. [07]
(b) Explain DECCA. [07]
- Q - 6 (a) Explain Operations of DME. [07]
(b) Define Following terms (4 and 5 each carry two marks). [07]
- 1) Doppler effect
 - 2) Janus X
 - 3) Drift Angle
 - 4) Track Stabilized antenna system
 - 5) Heading stabilized antenna system

OR

- Q - 6 (a) Explain the Pulse Doppler radar with neat block diagram. List the advantages of it's over CW radar. [07]
- (b) Write A short note on [07]
- 1) SLAR Radar
 - 2) Airport Radar
